

EHR Analytics and Cardiac Phenotyping from Echo Notes

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Abstract—Electronic health record (EHR) data contain rich clinical information but are often fragmented across structured tables and unstructured narrative reports, limiting their direct use for research and downstream analysis. In this project, we constructed a patient-level, analysis-ready dataset by integrating symptom-onset records with a large REDCap-derived clinical database containing laboratory measurements, vital signs, and echocardiographic results. Patient records were aligned at the hospitalization level using admission and discharge timestamps to ensure temporal consistency across measurements. Laboratory and vital sign data were preserved with exact collection timestamps and restructured into a wide-format representation to support patient-level analysis. Additionally, a rule-based natural language processing (NLP) pipeline was developed to extract clinically meaningful cardiac phenotypes, including left ventricular ejection fraction (LVEF), wall motion abnormalities, valvular abnormalities, and systolic and diastolic dysfunction, from free-text echocardiogram narratives. The resulting dataset captures both structured and narrative-derived clinical features while maintaining temporal fidelity, enabling exploratory analysis of cardiac function and physiological measurements. This work demonstrates a reproducible approach for transforming heterogeneous EHR data into structured formats suitable for clinical research and statistical modeling.

I. INTRODUCTION

Real-world EHR data present significant challenges for clinical research due to their complexity, heterogeneity, and lack of standardized structure. Patients may experience multiple hospitalizations, undergo repeated laboratory testing, and have diagnostic information recorded across both structured fields and unstructured clinical narratives. While structured data such as laboratory values and vital signs are relatively straightforward to extract, key diagnostic insights are often embedded within free-text reports, particularly in imaging studies such as echocardiograms.

Echocardiographic reports are a prime example of this challenge. Although they contain critical information about cardiac structure and function, including left ventricular ejection fraction and the presence of wall motion or valvular abnormalities, these findings are frequently described using narrative language rather than discrete variables. As a result, important clinical features are not directly accessible for quantitative analysis without additional processing.

Furthermore, temporal alignment is essential when working with EHR data. Laboratory values, vitals, and imaging results must be interpreted in the context of a specific hospitalization or clinical episode. Without careful alignment to admission and discharge times, analyses risk combining measurements from different clinical contexts, leading to incorrect conclusions.

The objective of this project was to address these challenges by constructing a unified, patient-level dataset that integrates symptom-onset information with temporally aligned laboratory, vital sign, and echocardiographic data. In addition, we aimed to extract structured cardiac phenotypes from unstructured echocardiogram narratives using NLP techniques, enabling exploratory analysis of cardiac function in relation to other physiological measurements.

II. DATASET OVERVIEW

This study utilized two primary data sources: a REDCap-derived clinical dataset and a symptom-onset analysis dataset. The REDCap dataset contains patient-level clinical information collected during hospital encounters, including demographic identifiers, admission and discharge timestamps, laboratory measurements, vital signs, procedural data, and imaging results. The symptom-onset dataset provides additional temporal context related to the onset of patient symptoms and serves as a reference cohort for patient inclusion.

The combined dataset includes thousands of unique patients and spans multiple years of clinical encounters. Laboratory data are recorded at high frequency, with many patients having repeated measurements for common labs such as sodium, potassium, calcium, creatinine, and troponin. Vital signs, including heart rate and blood pressure, are recorded with their respective measurement timestamps. Echocardiographic data consist of both structured fields and narrative reports, with the latter containing detailed clinical interpretations written by providers.

Due to the repeated and longitudinal nature of these measurements, the raw data are inherently long-format and episode-based. For analytical purposes, a patient-level wide-format representation is required, where each patient occupies a single row and relevant clinical features are organized into columns while preserving measurement timestamps.

III. METHODOLOGY

A. Patient and Hospitalization Matching

Patients were matched across datasets using a combination of patient name and medical record number (MRN). Hospitalization episodes were defined using admission and discharge timestamps from the clinical dataset. For patients with multiple admissions, each episode was treated separately during intermediate processing to ensure accurate temporal alignment of laboratory and vital sign measurements.

Only records corresponding to patients present in the symptom-onset dataset were retained for downstream analysis, ensuring cohort consistency across all data sources.

B. Laboratory Data Processing

Laboratory measurements were extracted from the RED-Cap laboratory sheet, which includes the laboratory name, result, unit, and exact datetime of collection (`Lab_Dttm`). To preserve clinical timing, all laboratory timestamps were retained and no narrative timestamps were used as proxies for lab collection times.

For each patient and hospitalization episode, laboratory measurements were filtered to those occurring within the admission–discharge window. For most laboratory tests, the first valid measurement within the hospitalization was selected to represent baseline laboratory status. This approach reduces redundancy while maintaining clinical relevance. Laboratory data were then reshaped into a wide format, where each laboratory variable is paired with its corresponding datetime, allowing direct comparison across patients.

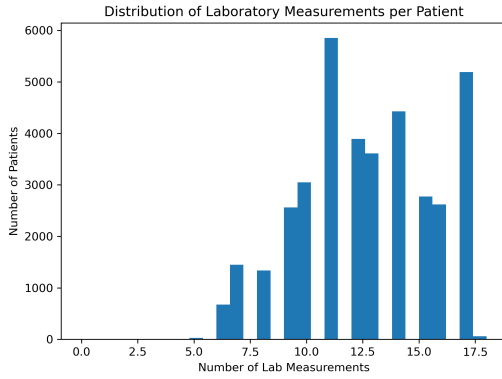


Fig. 1: Lab Measurements per Patient

Figure 1 illustrates the distribution of the number of laboratory measurements per patient, highlighting substantial variability in laboratory testing intensity across the cohort.

C. Vital Signs Integration

Vital sign data, including heart rate and blood pressure, were integrated using their respective measurement timestamps (`Recorded_Dttm` for heart rate and `Service_Dttm` for blood pressure). Similar to laboratory data, vital sign measurements were aligned to hospitalization episodes and preserved with their original timestamps. This ensured that all physiological measurements corresponded to the correct clinical context.

D. Electrocardiogram NLP Pipeline

Echocardiographic narrative reports were processed using a rule-based NLP pipeline designed to extract clinically meaningful cardiac features. Narrative-only records were grouped by patient, exam datetime, and accession number to reconstruct complete echocardiogram reports.

The NLP pipeline extracted the following features

- **Left ventricular ejection fraction** (numeric values and qualitative descriptors)
- **Wall motion abnormalities**
- **Valvular abnormalities**

- **Systolic dysfunction**
- **Diastolic dysfunction**

The extraction logic accounted for common clinical language patterns, including negations (e.g., “no”, “without”), severity descriptors (e.g., mild, moderate, severe), and variations in terminology (e.g., “abnormal”, “abnormalities”). Extracted features were stored separately to allow for manual validation and quality assessment.

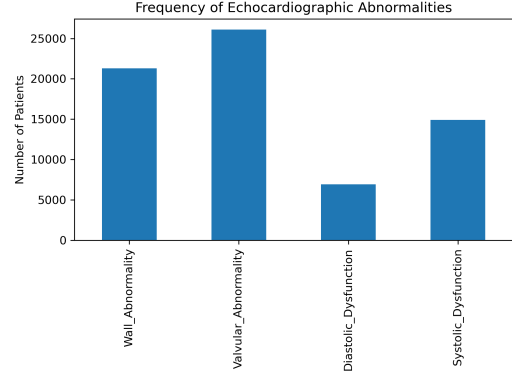


Fig. 2: Echo abnormalities

Figure 2 summarizes the frequency of echocardiographic abnormalities identified through the NLP process.

IV. RESULTS

A. Laboratory and Vital Sign Characteristics

Laboratory testing frequency varied widely across patients, as shown in Figure 1, reflecting differences in clinical complexity and hospitalization duration. Most patients underwent multiple laboratory tests during admission, underscoring the importance of systematic data reduction and temporal alignment. Vital sign measurements exhibited similar variability but were consistently aligned to hospitalization windows using their respective timestamps.

B. Cardiac Function Assessment

The distribution of left ventricular ejection fraction values extracted from echocardiogram narratives is shown in Figure 3. The distribution spans reduced, borderline, and preserved ejection fraction ranges, demonstrating the ability of the NLP pipeline to capture clinically meaningful cardiac function metrics from free-text reports.

The relationship between LVEF and heart rate is illustrated in Figure 4. While no strong linear association is observed, the scatter plot provides exploratory insight into physiological variability across patients with differing levels of cardiac function.

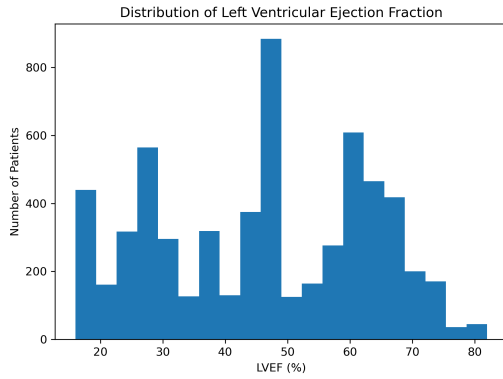


Fig. 3: LVEF Distribution

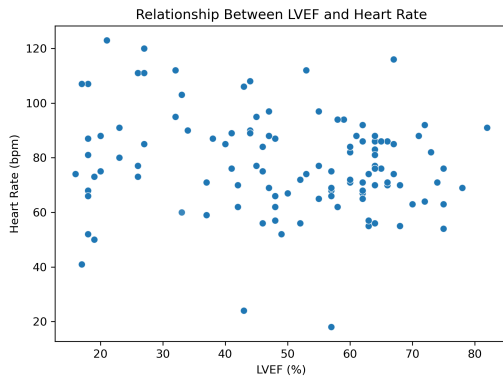


Fig. 4: Relationship between LVEF and heart rate, demonstrating clinically interpretable trends.

C. Anthropometric and Correlation Analysis

Additional exploratory analyses were conducted to examine relationships between anthropometric variables and cardiac measures. Figure 5 presents the distribution of body mass index (BMI) across the cohort, showing a concentration in the overweight and obese ranges.

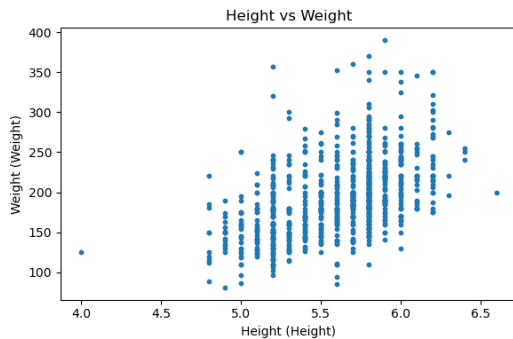


Fig. 5: BMI Relationship

A correlation heatmap further illustrates pairwise relationships between BMI, blood pressure, heart rate, and LVEF, revealing modest correlations consistent with known physiological relationships.

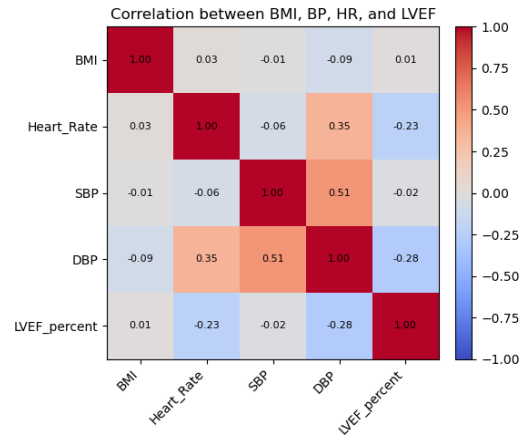


Fig. 6: Correlation Heatmap

V. REFLECTIONS, LEARNINGS AND FUTURE DIRECTIONS

This project highlighted the complexity and ambiguity inherent in real-world clinical data and reinforced the importance of rigorous data engineering practices in healthcare analytics. One of the primary learnings was that meaningful clinical insights are often embedded in unstructured narrative text rather than clean, structured fields. Extracting these insights required careful handling of clinical language, including negation, severity modifiers, and inconsistent phrasing across providers. Developing and iteratively refining the rule-based NLP pipeline emphasized the need for domain-aware logic rather than naive keyword matching when working with medical narratives.

Another key learning involved the critical role of temporal alignment in EHR analysis. Laboratory values, vital signs, and imaging results must be interpreted within the correct hospitalization context, and even small misalignments in timestamps can substantially alter analytical outcomes. Preserving original measurement datetimes and explicitly aligning data to admission and discharge windows proved essential for maintaining clinical validity.

From a data management perspective, this project underscored the trade-offs between completeness and usability. While EHR systems capture extensive longitudinal data, transforming these records into a patient-level, analysis-ready format requires deliberate decisions about aggregation, feature selection, and representation. These decisions must balance analytical tractability with clinical relevance.

Future work could extend this pipeline by incorporating more advanced NLP techniques, such as transformer-based models, to capture subtler clinical nuances and reduce reliance on handcrafted rules. Additionally, expanding the temporal analysis to model longitudinal trends across multiple hospitalizations may provide deeper insight into disease progression and patient outcomes.